

Open source cloud platforms are built on principles of open collaboration and transparency. They allow users to access, modify, and distribute the source code, which facilitates customisation and ensures that the platforms evolve to meet diverse needs. This approach contrasts with proprietary platforms, where users are often limited to the features and capabilities provided by the vendor.

Benefits of open source cloud platforms

Cost savings: Open source platforms eliminate the licensing fees associated with proprietary software. Organisations can deploy these solutions with minimal upfront costs and pay only for the resources they use.

Flexibility and customisation: Users can modify the source code to tailor the platform to their specific requirements. This flexibility enables the integration of new features and optimisations that align with unique business needs.

Community support: A vibrant community of developers and users contributes to the ongoing improvement and support of open source cloud platforms. This community-driven approach ensures that issues are addressed quickly and innovations are incorporated into the platform.

Examples of open source cloud platforms

OpenStack: This leading open source cloud platform is designed for building and managing cloud infrastructure. It provides a comprehensive suite of services, including compute, storage, and networking, allowing users to create private or public clouds.

Apache CloudStack: An open source cloud computing platform for creating and managing large-scale cloud environments, it offers features for compute, storage, and network management, with a focus on ease of use and scalability.

OpenNebula: A flexible open source platform for building and managing private, public, and hybrid clouds, it provides a unified view of cloud resources and supports various virtualisation technologies.

Serverless computing and open source

Serverless computing represents a transformative shift in cloud architecture, enabling developers to build and run applications without managing server infrastructure. In this model, the cloud provider handles server provisioning, scaling, and maintenance, allowing developers to focus solely on writing code.

Serverless computing allows applications to be broken down into individual functions or services that execute in response to events or triggers. These functions run in stateless, ephemeral environments, scaling automatically based on demand. This model contrasts with traditional server-based computing, where developers must manage servers, handle scaling, and ensure high availability.

The benefits of serverless computing are:

Reduced operational overhead: Developers are relieved from the burden of managing servers and infrastructure, leading to faster development cycles and reduced operational complexity.

Automatic scaling: Serverless platforms automatically scale resources based on the volume of incoming requests, ensuring optimal performance without manual intervention.

Cost efficiency: Users pay only for the compute time consumed by their functions, which can result in significant cost savings compared to provisioning dedicated servers.

Imagine a company that needs to process user-uploaded images in real-time. With serverless computing, it can deploy a function that is triggered whenever a new image is uploaded. The function processes the image (e.g.,

resizing, filtering) and stores the result in a cloud storage service. This approach eliminates the need for dedicated servers and scales automatically based on the volume of image uploads.

Open source serverless platforms

The rise of serverless computing has also been embraced by the open source community, leading to the development of various open source serverless platforms that offer benefits similar to proprietary solutions.

Apache OpenWhisk: Apache OpenWhisk is an open source serverless platform that provides a framework for building and deploying serverless applications. It supports multiple programming languages and allows developers to define functions, triggers, and actions.

Kubeless: Kubeless is another open source serverless platform designed to run on Kubernetes. It integrates with Kubernetes to manage serverless functions and offers features such as automatic scaling and event-driven execution.

Cloud security and open source

As organisations increasingly adopt cloud computing, security concerns become paramount. Cloud environments are susceptible to various threats, including data breaches, misconfigured access controls, and unauthorised access. Open source tools play a crucial role in addressing these challenges by offering transparency, customisation, and robust security features.

Common security threats are:

Data breaches: Unauthorised access to sensitive data can occur due to weak security practices or vulnerabilities in cloud infrastructure.

Misconfigured access controls: Incorrectly configured permissions and access controls can expose data and services to unauthorised users.

Insider threats: Employees or contractors with access to cloud